A PROJECT TITLE

ON

Diabetes Risk Prediction System using Machine Learning Classification Algorithms

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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Name of Title: Diabetes Risk Prediction System using Machine Learning Classification Algorithms

1. Abstract:

Diabetes is a chronic disease that affects the body's ability to regulate blood glucose levels and can lead to serious health complications if it is not identified and managed at an early stage. This project presents **a** Diabetes Risk Prediction System using Machine Learning Classification Algorithms to predict whether a person is likely to belong to a diabetes-positive or diabetes-negative class based on patient information and reported symptoms.

The proposed system uses a diabetes dataset containing 520 patient records and 17 columns. The dataset includes attributes related to age, gender, and various diabetes-associated symptoms such as polyuria, polydipsia, sudden weight loss, weakness, polyphagia, genital thrush, visual blurring, delayed healing, and muscle stiffness. Since machine learning algorithms require numerical input, categorical values such as Yes/No and Male/Female are converted into numerical representations during preprocessing.

Two supervised machine learning algorithms, Logistic Regression and Random Forest Classifier, are implemented for diabetes classification. The dataset is divided into training and testing sets, with 80% of the data used for training and 20% used for testing. The models are evaluated using accuracy, confusion matrix, classification report, and five-fold cross-validation. Logistic Regression achieved an average cross-validation accuracy of 87.5%, while Random Forest achieved an average accuracy of 91.54%. Therefore, Random Forest is selected as the best-performing model for the proposed system.

The system also provides an interactive facility for entering details of a new patient. The selected Random Forest model processes the entered information and predicts whether the patient is classified as diabetes-positive or diabetes-negative. The proposed system demonstrates how machine learning can be applied to healthcare-related classification tasks and can serve as a supportive tool for preliminary diabetes risk prediction. The prediction should not be considered a medical diagnosis and should be confirmed by appropriate clinical evaluation.

2. INTRODUCTION:

Diabetes is one of the major health conditions affecting people worldwide. Early identification of diabetes risk can help individuals seek appropriate medical evaluation and take preventive measures. Traditional diagnosis depends on clinical examination and medical tests, while machine learning provides an additional approach for analyzing patient information and identifying patterns associated with diabetes.

The main objective of this project is to develop a machine learning-based system that can classify a patient as diabetes-positive or diabetes-negative using demographic information and reported symptoms. The system uses a diabetes dataset containing 520 records. The categorical data is converted into numerical values using Label Encoding so that it can be processed by machine learning algorithms.

Two classification algorithms are used in this project: Logistic Regression and Random Forest. Logistic Regression provides a simple classification approach, while Random Forest combines multiple decision trees to make predictions. Both models are trained and evaluated, and five-fold cross-validation is used to compare their general performance.

The Random Forest model achieved an average cross-validation accuracy of 91.54%, compared with 87.5% for Logistic Regression. Based on this comparison, Random Forest is selected for predicting new patient cases.

The developed system accepts patient details through user input and generates a classification result. The project demonstrates the application of machine learning in healthcare data analysis and provides a simple framework for preliminary diabetes risk prediction.

3. SOFTWARE REQUIREMENTS:

3.1 Programming Language

Python is used as the primary programming language for implementing the diabetes prediction system.

3.2 Libraries and Frameworks

The following Python libraries are used:

- **Pandas** – for loading, processing, and manipulating the dataset.
- **Scikit-learn** – for machine learning algorithms, preprocessing, model evaluation, and cross-validation.
- **LabelEncoder** – for converting categorical data into numerical values.
- **LogisticRegression** – for implementing the Logistic Regression classification algorithm.
- **RandomForestClassifier** – for implementing the Random Forest classification algorithm.
- **train_test_split** – for dividing the dataset into training and testing sets.
- **cross_val_score** – for performing five-fold cross-validation.
- **accuracy_score** – for calculating model accuracy.
- **confusion_matrix** – for evaluating correct and incorrect predictions.
- **classification_report** – for calculating precision, recall, F1-score, and support.

3.3 Dataset

The project uses the dataset:

diabetes_data_upload.csv

The dataset contains:

- **520 records**
- **17 columns**
- Patient demographic and symptom-related attributes
- A **class** column representing the target output

3.4 Development Environment

The project can be developed and executed using:

- Python 3.x
- Jupyter Notebook or Python IDE
- Windows/Linux/macOS operating system
- Minimum 4 GB RAM recommended
- Python package manager such as pip

3.5 Machine Learning Algorithms

The project implements:

1. **Logistic Regression**
2. **Random Forest Classifier**

The models are evaluated using testing accuracy, confusion matrix, classification report, and five-fold cross-validation.

Signature of the Guide

Course Instructor